

REQUIREMENT ANALYSIS

Language Learning Hub- Flashcard CRUD, AI Generation, Sharing & Export

Testing Portfolio Project

Github: [LLH-Functional-and-API-Testing-Project](#)

Github Web Project: [Language-learning-hub-for-quizz](#)

Url: [link web](#)

Requirement Basis: BRD.md v1.9 (BR-02) and FRD-DOC.md v0.5 (Group 3: FR-DOC-009–012; Group 4: FR-DOC-013–015; Group 5: FR-DOC-016–019), incorporated with Phase 2 Changes (CR-DOC-001, CR-DOC-002, CR-CROSS-001) and Production Hotfixes (DEF-AI-01–04, DEF-UI-01–02, DEF-SAVE-01).

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1. Overview & Scope

- **Business requirement:** BR-02 - Managing Document.
- **Business purpose:** Let a learner create and manage high-quality vocabulary material in the hierarchy Folder > Deck > Flashcard, accelerate preparation with reviewable AI generation, safely expose a selected Deck for read-only public viewing, and take an offline Excel copy of its current Flashcards.
- **Actors:**
 - **Learner:** Authenticated Active owner of the Deck and its Flashcards.
 - **Public User:** Unauthenticated visitor who opens an active shared Deck URL; read-only.
 - **AI Service (Groq):** External service that generates temporary Flashcard preview content from valid vocabulary input.
 - **System:** Enforces session, account status, ownership, public-link state, and downstream consistency.
- **Module preconditions:** The system is operational; Learner session is valid; Learner account is Active; every Learner action is limited to resources they own. FR-DOC-017 is the sole public-access exception.

1.1 In scope

- View the Flashcards of an owned Deck, including the empty-list state.
- Manually create, edit, and permanently delete an owned Flashcard.
- Validate Flashcard content, language placement guidance, duplicate terms, and the non-movable Deck association.
- Initialize learning state for a newly saved manual Flashcard and remove/invalidate it when deleted.
- Submit a vocabulary list to AI Service, validate and review a temporary generated preview, and save selected valid generated Flashcards to an existing Deck or a new Deck under an owned Folder.
- Enable/disable a public share link for an owned Deck; view the active link as a Public User.
- Export the current contents of a non-empty owned Deck as .xlsx.
- RBAC, ownership checks, shared-link revocation, privacy of public data, and effects on LEARN/DASH.

1.2 Out of scope

- Folder/Deck CRUD(Groups 1, 2).
- Creating a Deck without an existing Folder; moving Flashcards between Decks.
- Editing, exporting, importing/copying, learning from, or collaboratively editing a public shared Deck.

- Guaranteeing content accuracy; detailed token algorithm, endpoint/database design, and implementation-specific Excel generation.

2. Business Requirements

- **BR-02.1 - Ownership and hierarchy:** A Flashcard belongs to exactly one Deck. A Learner may only manage their own content.
- **BR-02.2 - Flashcard completeness:** term, meaning, and exampleSentence are mandatory, distinct content elements and must be non-empty after trimming before save.
- **BR-02.3 - Language content placement:** Japanese and Chinese content follow their respective placement rules; a phonetic reading that is inapplicable or unavailable does not block saving.
- **BR-02.4 - Flashcard uniqueness and immobility:** A normalized term is unique within one Deck; editing cannot move a Flashcard to another Deck.
- **BR-02.5 - Learning-data integrity:** A newly saved manual Flashcard is initialized as new/unreviewed. A deleted Flashcard must immediately cease to feed future learning, dashboard, shared-view, and export retrievals.
- **BR-02.5A - AI input and preview boundary:** The system parses, normalizes, deduplicates, and limits AI vocabulary input; generated output is temporary, reviewable, and never becomes learning content before explicit Learner confirmation.
- **BR-02.5B - AI partial-result safety:** Valid generated items may be previewed and saved when other items fail, but missing/invalid generated items cannot be selected for save.
- **BR-02.5C - AI save consistency:** Saved AI-generated Flashcards use the same three approved fields, ownership/duplicate validation, and initial SRS state as manually created Flashcards.
- **BR-02.6 - Controlled public sharing:** One owned Deck may have a unique, unguessable active URL. Public access is anonymous, read-only, and exposes only approved content.
- **BR-02.7 - Immediate revocation:** Disabling sharing or deleting the Deck makes its URL inactive. Re-enabling creates a new token; the old one remains invalid.
- **BR-02.8 - Non-mutating export:** An owned non-empty Deck can be exported in the exact .xlsx content format; sharing/export do not change Flashcards, SRS, sessions, or dashboard metrics.

3. Functional Requirements & Logic Flow

3.1 Functional requirements and acceptance baseline

FR	Description	Acceptance baseline / exception coverage
FR-DOC-009	Learner views Flashcards in an owned Deck.	Show only current Flashcards; show “This Deck has no Flashcards yet.” for an empty Deck; deny cross-owner access; handle missing Deck as Not Found.
FR-DOC-010	Learner manually creates a Flashcard.	Save valid fields in target Deck; trim and reject blank/over-limit fields; reject normalized duplicate term in the same Deck; initialize reviewLevel=0, nextReview=null, lastRecallRating=null; deny missing/cross-owner Deck.
FR-DOC-011	Learner edits an owned Flashcard.	Update valid content only; preserve its Deck association and existing learning state; keep prior data if validation/duplicate check fails; reject a movement attempt, Not Found, or cross-owner request.
FR-DOC-012	Learner deletes an owned Flashcard.	Permanently delete without confirmation; remove/invalidate associated SRS state; no other Flashcard changes on failure; exclude the deleted item from downstream retrieval.
FR-DOC-013	Learner submits a vocabulary list for AI generation.	Learner selects mandatory targetLanguage (Japanese or Chinese) and submits vocabulary list. Parse comma/newline/mixed input; trim, ignore blanks, deduplicate, accept at most 50 items and 3,000 characters; reject request with 400 Bad Request if target language is missing or unsupported; proceed with valid items for the selected target language where possible; do not save a Flashcard.
FR-DOC-014	System generates an AI Flashcard preview.	Validate required three fields, target language mapping, Sino-Vietnamese reading rules (required for Chinese Hanzi and Japanese Kanji; null for Hiragana/Katakana-only; Title Case capitalization), and contextual example sentence with Vietnamese translation. Display valid and "uncertain but safe to review" items in temporary preview; surface missing/invalid/wrong-language/timed-out generation in failedItems with specific failure reasons without persistence.
FR-DOC-015	Learner reviews and saves AI-generated Flashcards.	Permit edits and selection; when an Existing Deck is selected, visually identify duplicate normalized terms on UI before save (recalculated on term edit, Deck change, or selection change); prevent save while duplicate selected items exist; save valid non-duplicate selected items to an owned existing Deck or a

		valid new Deck under an owned Folder; validate duplicate/ownership/required fields and initialize SRS state.
FR-DOC-016	Learner enables public sharing for an owned Deck.	Set sharing to enabled and create an active URL /share/deck/{shareToken}; if already enabled, show the existing URL; deny missing/cross-owner Deck.
FR-DOC-017	Public User views an active shared Deck.	Show only deckName, term, meaning, exampleSentence in view-only mode; return unavailable state for invalid, disabled, or deleted link; display empty state for an empty Deck; deny restricted actions.
FR-DOC-018	Learner disables public sharing.	Set sharing disabled and immediately revoke prior link; a refresh/next request from an open public page must be unavailable; re-enable creates a new token; deny cross-owner request.
FR-DOC-019	Learner exports an owned Deck.	Produce .xlsx only for a non-empty owned Deck; exact columns/order and UTC+7 filename; retain supported multiline content; deny public/cross-owner use; do not alter application data on export failure.

4. Field Specifications & Data Constraints

Field / output	Control or type	Required	Validation / constraint	Default / state	Dependencies / notes
term	Multiline text input / text	Yes	Non-empty after trim; max 500 chars; line breaks allowed; normalized (trim, collapse internal spaces, case-insensitive where applicable) must be unique in the Deck. Japanese: original vocabulary + applicable Sino-Vietnamese (required when vocabulary contains	None	Shown in full on Flashcard front and shared view/export.

			Kanji; null for Hiragana/Katakana-only) . Chinese: original vocabulary only. Sino-Vietnamese syllables use Title Case capitalization. UI renders multiline term in a multiline editor control, preserving line breaks.		
meaning	Multiline text input / text	Yes	Non-empty after trim; max 1,000 chars; line breaks allowed; must be conceptually separate from example. Japanese: applicable Hiragana then Vietnamese meaning. Chinese: Sino-Vietnamese, optional Pinyin, then Vietnamese meaning. Vietnamese meaning must never be placed in the Sino-Vietnamese reading line.	None	Absence of unavailable/inapplicable Hiragana or Pinyin does not block manual save. Shown on back/shared view/export.
example Sentence	Multiline text input / text	Yes	Non-empty after trim; max 500 chars; line breaks allowed; target-language contextual sentence followed by Vietnamese translation. Missing target sentence, missing Vietnamese translation, or wrong target language invalidates preview item.	None	Shown separately on back/shared view/export.
flashcardDeck Association	Immutable relationship	Yes	Exactly one Deck; changing it is out of MVP scope.	Target Deck at create	Must remain unchanged on edit.

review Level	Number	System	Initialized to 0 when manual Flashcard is saved.	0	Used by LEARN/DASH; no recalculation on content edit.
nextReview	Date/null	System	Initialized null; null means new/unreviewed.	null	Makes saved item eligible for Study Today.
lastRecallRating	Enum/null	System	Initialized null.	null	SRS data is removed/invalidated on deletion.
vocabularyList	Multiline text input / collection	Yes for AI request	One item per line, comma-separated, or mixed; trim items; ignore empty lines/extra spaces; deduplicate normalized terms; max 50 items and 3,000 characters; requires explicit targetLanguage (Japanese or Chinese).	None	Unsupported items or items inconsistent with target language are returned in failedItems; missing targetLanguage rejects request before provider call.
generationStatus	System enum	System	pending, completed, partial, failed, or timedOut.	pending after valid submission	timedOut/failed does not save content.
generatedFlashcardPreview	Temporary editable collection	Conditional	Each item requires valid term, meaning, exampleSentence and approved language mapping before it is save-eligible. UI layout: Term and Meaning match in width/height; Example sentence occupies full row below.	Not persisted	Discarded on cancel, refresh, logout, or session expiry; excluded from LEARN, DASH, sharing, and export until saved.
AI save target	Existing Deck selector	Yes when	Existing target must be owned and exist; new Deck requires an owned	None	Duplicate generated terms in Existing Deck

	/ new Folder + deckName	saving	existing Folder plus valid non-duplicate deckName.		are pre-checked and identified on UI before save (DEF-SAVE-01). Existing Chinese data policy: old-format Chinese Flashcards are not auto-migrated; editing an old card may return 409 if a new-format term already exists.
sharingStatus	Toggle / enum	System	enabled or disabled; only owner may change it.	disabled assumed for a new Deck; confirm in AMB-01	Affects Public User access only.
shareToken	System-generated opaque token	Conditional	Unique, unguessable; no readable learner/deck/email/private data; not reused after revoke.	Created on enable	Technical algorithm intentionally unspecified.
sharedDeckUrl	Read-only URL	Conditional	/share/deck/{shareToken}; active only while sharing enabled and Deck exists.	Created on enable	Does not auto-expire.
exportFile	.xlsx download	Conditional	Only non-empty owned Deck. Filename deck_<normalizedDeckName>_<YYYYMMDD>.xlsx, UTC+7.	Generated on request	normalizedDeckName: lowercase, spaces to hyphens, unsafe characters

					removed/replaced.
Export columns	Ordered worksheet columns	Yes when export succeeds	Exact order: No, Term, Meaning, Example Sentence; preserve multiline where supported.	N/A	Excludes private/system/SRS information.

5. RBAC & Downstream Impacts

5.1 Permission matrix

Action / resource	Learner - owner	Learner - non-owner	Public User	Admin
View Flashcards in a Deck	Allowed	Denied	Denied except active shared view	Denied
Create / edit / delete Flashcard	Allowed	Denied	Denied	Denied
Submit AI vocabulary / view own preview	Allowed	Denied	Denied	Denied
Edit/select/save own generated preview	Allowed	Denied	Denied	Denied
Generate preview content	System invokes Groq for owner request	N/A	Denied	Denied
Enable / disable sharing	Allowed	Denied	Denied	Denied
View active shared link	May view	May view as public content only	Allowed, view-only	May view as public content only

Export Deck	Allowed for non-empty Deck	Denied	Denied	Denied
View owner/SRS/dashboard/internal data through share page	N/A to public page	N/A	Denied / not exposed	N/A

5.2 Downstream impacts and dependencies

Change/event	Affected area	Required impact
Manual Flashcard saved	LEARN, DASH	Initialize new learning state; item becomes eligible for Study Today. Toast UI feedback; does NOT advance profiles.last_learning_activity_at and creates no notification row.
AI preview generated but not saved	DOC, LEARN, DASH, sharing, export	Temporary only; it must not affect learning queues, dashboard metrics, public view, or export.
Valid generated Flashcards saved	DOC, LEARN, DASH, sharing, export	Persist as ordinary Flashcards, initialize reviewLevel = 0, nextReview = null, lastRecallRating = null, and make them available to downstream modules. Toast UI feedback; does NOT advance profiles.last_learning_activity_at and creates no notification row.
Flashcard content edited	DOC, LEARN/DASH	Content updates; Deck association and learning state remain unchanged. Toast UI feedback; no activity/notification record.
Flashcard deleted	LEARN, DASH, public view, export	Immediately exclude it from future Study Today/Freedom Mode queues, active/future LEARN retrieval, dashboard source calculations, shared Deck retrieval, and export. Toast UI feedback; no activity/notification record.
Deck sharing enabled/disabled	Public shared-view access	Only changes public access; does not change learning eligibility, SRS, DASH, or the owner's Freedom Mode availability. Toast UI feedback; no activity/notification record.

Deck deleted	Shared link	Invalidate its public URL.
Deck exported	DOC, LEARN, DASH, sharing	Read-only operation; no state/content/metrics/sharing change.

6. Traceability Matrix

BR / rule	Functional requirement	Acceptance criteria covered	Verification focus
BR-02.1; BR-DOC-008	FR-DOC-009	AC-DOC-009-01 to 03	Owned list, empty state, ownership.
BR-02.2, BR-02.3; BR-DOC-008, 009, 015, 020	FR-DOC-010	AC-DOC-010-01 to 05	Required/boundary fields, normalized duplicate, language placement, initial SRS state, RBAC.
BR-02.2–02.4; BR-DOC-008, 009, 010, 015	FR-DOC-011	AC-DOC-011-01 to 06	Update integrity, validation rollback, duplicate exclusion, immutable Deck relationship, RBAC.
BR-02.5; BR-DOC-011, 020	FR-DOC-012	AC-DOC-012-01 to 04	Permanent deletion and downstream exclusion.
BR-02.5A; BR-DOC-012, 013, 015	FR-DOC-013	AC-DOC-013-01 to 08	Input parsing, normalization, limits, supported-item handling, no persistence, provider timeout.
BR-02.3, BR-02.5A–B; BR-DOC-013, 014, 015	FR-DOC-014	AC-DOC-014-01 to 04	Required fields, language mapping, partial results, invalid-item exclusion, provider failure.
BR-02.2–02.5C; BR-DOC-004–006, 008–009, 013–015, 020	FR-DOC-015	AC-DOC-015-01 to 08	Editable preview, target ownership, new/existing Deck, duplicate/validation controls,

			discard boundary, initial SRS state.
BR-02.6–02.8; BR-DOC-016–018, 021	FR-DOC-01 6	AC-DOC-0 16-01 to 04	Token/URL activation, idempotency, ownership.
BR-02.6; BR-DOC-016–018, 021	FR-DOC-01 7	AC-DOC-0 17-01 to 06	Active-token validation, public data minimization, view-only enforcement.
BR-02.7; BR-DOC-016–018, 021	FR-DOC-01 8	AC-DOC-0 18-01 to 05	Immediate revocation, refresh behavior, old-token non-reuse.
BR-02.8; BR-DOC-019, 021	FR-DOC-01 9	AC-DOC-0 19-01 to 07	File type/content/order/name, empty/unauthorized/error paths, non-mutation.